The ProbLemma's Channel How To

An Answer To The Question "How Do I Use This Channel"?

The Fundamentals

I. Watch Season 1 Episode 1.

Why. In that episode we explain what Mathematics is, what Mathematics is not and what is not Mathematics.

TLDR: Mathematics is an art of telling a story of how and why … wrapped into the technical language of logical connectives (NOT, AND, OR), universal quantifiers (for all), existential quantifiers (exists) and IF P THEN Q implications. If you are not telling a story then you are not doing Mathematics.

II. Watch Season 1 Episode 2.

Why. In that episode we explain how to practice Mathematics correctly because right now you are doing Mathematics incorrectly.

TLDR: always try to solve all the problems and prove all the theorems that we post on this channel completely by yourself and solely on your own first. If you fail to do so then do not watch our presentation in its entirety but only watch as much of the material as will make you go "Aha! I see what is going on here".

At which point we recommend that you walk away from the computer and complete the solution of the problem at hand on your own. Repeat the last two steps until you can solve the problem completely, on your own and in one continuous motion.

Keep good records of your attempts. In 6 to 12 months of sustained effort you will learn a lot about yourself, after you go over these records.

The above two episodes can also be found in "The Fundamentals" playlist.

Basic Training

Work your way through the following, 11, basic problem-solving approaches.

Why. Problem-solving approaches are heuristic templates that travel far and wide not only throughout Mathematics but throughout Physics and Computer Science and beyond.

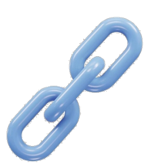
TLDR: these heuristic templates, which are short, sweet and local professional jargon-free in their formulation, provide a high-level tactical guidance of what to try.

- Transform And Conquer (6 episodes)
- Reinterpret And Conquer (4 episodes)
- Reverse Order (8 episodes)
- Space-Time (4 episodes)
- Equation (5 episodes)
- Scope Reduction (4 episodes)
- Scope Expansion (5 episodes)
- Eliminate And Conquer (6 episodes)
- Divide And Conquer (7 episodes)
- Invariant (6 episodes)
- Mathematical Equivalence (4 episodes)

for a total of 58 episodes. At the rate of 1 episode per day every day you will done with this in 2 months.

Enjoy Yourself

Begin solving problems in Mathematics, (Theoretical) Physics and Computer Science the right way, since you now know how. Ad infinitum. Pick a problem on this channel that resonates with you and enjoy both the problem and the time that you spend on the solution of that problem and yourself.



The menu of problem choices available to you is captured in the Season Guide free PDFs. In other words, all the problems, without exception, that we solve on this channel are indexed by episode name and problem name and number with the corresponding link.

The dedicated GitHub repository of this channel can be found here.

All the Season Guide PDFs can be found here.

In particular:

- the Season 1 Guide is here
- the Season 2 Guide is here
- the Season 3 Guide is here

Poke around that area of the repository for more recent season guides.

Content Partitioning

All the hard-science material that we publish on this channel can be partitioned into three big categories: Mathematics, Physics and Computer Science.

All episodes of our channel that pertain to Physics go into the "Physics" playlist, see below.

All the episodes that pertain to Computer Science go into the "Computer Science" playlist.

However, only some episodes on this channel that pertain to Mathematics are put into their dedicated playlists, also see the details of this sorting below.

Physics

All the episodes on this channel that pertain to **Physics** are gathered in one playlist named "Physics"

Computer Science

All the episodes on this channel that pertain to **Computer Science** are gathered in one playlist named "Computer Science"

Mathematics

All the episodes on this channel that pertain to **Integral Calculus** are gathered in one playlist named "Integrals Are Us"

In addition, all the integrals that we evaluate on this channel are indexed in the free "Integrals" PDF that can be found here.

All the episodes on this channel that pertain to **Sums**, both finite and infinite, are gathered in one playlist named "Sums Are Us"

All the sums that we evaluate on this channel are indexed in the free "Sums" PDF that can be found here.

All the episodes on this channel that pertain to **Limits Calculus** are gathered in one playlist named "Limits Are Us"

All the limits that we evaluate on this channel are indexed in the free "Limits" PDF that can be found here.

All the episodes on this channel that pertain to **Sums**, both finite and infinite, are gathered in one playlist named "Sums Are Us"

All the sums that we evaluate on this channel are indexed in the free "Sums" PDF that can be found here.

All the episodes on this channel that pertain to **Products**, both finite and infinite, are gathered in one playlist named "Products Are Us"

All the sums that we evaluate on this channel are indexed in the free "Products" PDF that can be found here.

Courses

The list of all the courses available on this channel can be found here.

For example, a course on elementary synthetic geometry-based introduction to Inversive Geometry can be found here.

Auxiliary Material

The auxiliary material, such as the source code listings, for example, can be found in my free GitHub repository here. Poke around there.

The names of the files stored in this part of the repository are quite suggestive and should be easy to correlate with the episode number and the type, the name and the number of the corresponding problem to which these files pertain.

Books

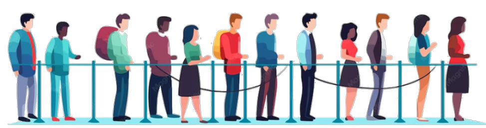
The extra curious and advanced travelers may be interested in my free books on mathematics and physics. All such books are located in my free GitHub repository here.

For example:

- an elementary introduction to the Theory Of Groups book can be found here and
- an elementary synthetic geometry-based introduction to Inversive Geometry book can be found here

Poke around for more goodies there.

Stay Informed



Get the heads-up on what is in the queue for the upcoming episodes on our channel by checking the "Posts" section of the channel here.

Keep in touch and tell me what you think about this channel.